

SAMPLE DOCUMENT - FICTITIOUS DATA FOR DEMONSTRATION ONLY

Mechanical Datasheet - Separator V-101

Contractor document MDS-V-101 Rev 1. Design code: ASME BPVC Section VIII, Division 1, 2025 edition.

Equipment No.	V-101
Orientation	Vertical
Inside diameter	1800 mm
Length, tangent-to-tangent	5400 mm
Vessel support	Skirt
Shell corrosion allowance internal	3 mm
Heads corrosion allowance internal	3 mm
Nozzles corrosion allowance internal	3 mm
Shell joint efficiency	1.0
Heads joint efficiency	1.0
PWHT	Per code
Insulated	Yes
Insulation by	Others
External insulation	Mineral wool, 50 mm
Insulation density	128 kg/m3
MAWP basis	Calculated by manufacturer

Materials

Shell	SA-516 Gr. 70N
2:1 ellipsoidal heads	SA-516 Gr. 70N
Nozzles	SA-106 Gr. B
Flanges	SA-105N
Bolting	SA-193 B7 / SA-194 2H
Reinforcing pads	SA-516 Gr. 70N
Internals	SS 316L
Attachments	SA-516 Gr. 70

Nozzle schedule

Mark	Service	Qty	Size	Flange type	Class
N1	Gas/liquid inlet	1	8 in	WNRF	300
N2	Gas outlet	1	6 in	WNRF	300
N3	Liquid outlet	1	3 in	WNRF	300
N4	PSV connection	1	4 in	WNRF	300
N5	Level instrument	2	2 in	WNRF	300

M1	Manway	1	24 in	WNRF	300
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Welded joints

Category A joint type	Type 1
Category A NDE	Full RT per UW-51
Category B head-to-shell joint type	Type 1
Category B head-to-shell NDE	Full RT per UW-51
Category D joint type	Full penetration
Category D NDE	MT after PWHT